AIR POLLUTION

Debasree Purkayastha
School of Infrastructure, IIT Bhubaneswar
Email id: debasreep@iitbbs.ac.in

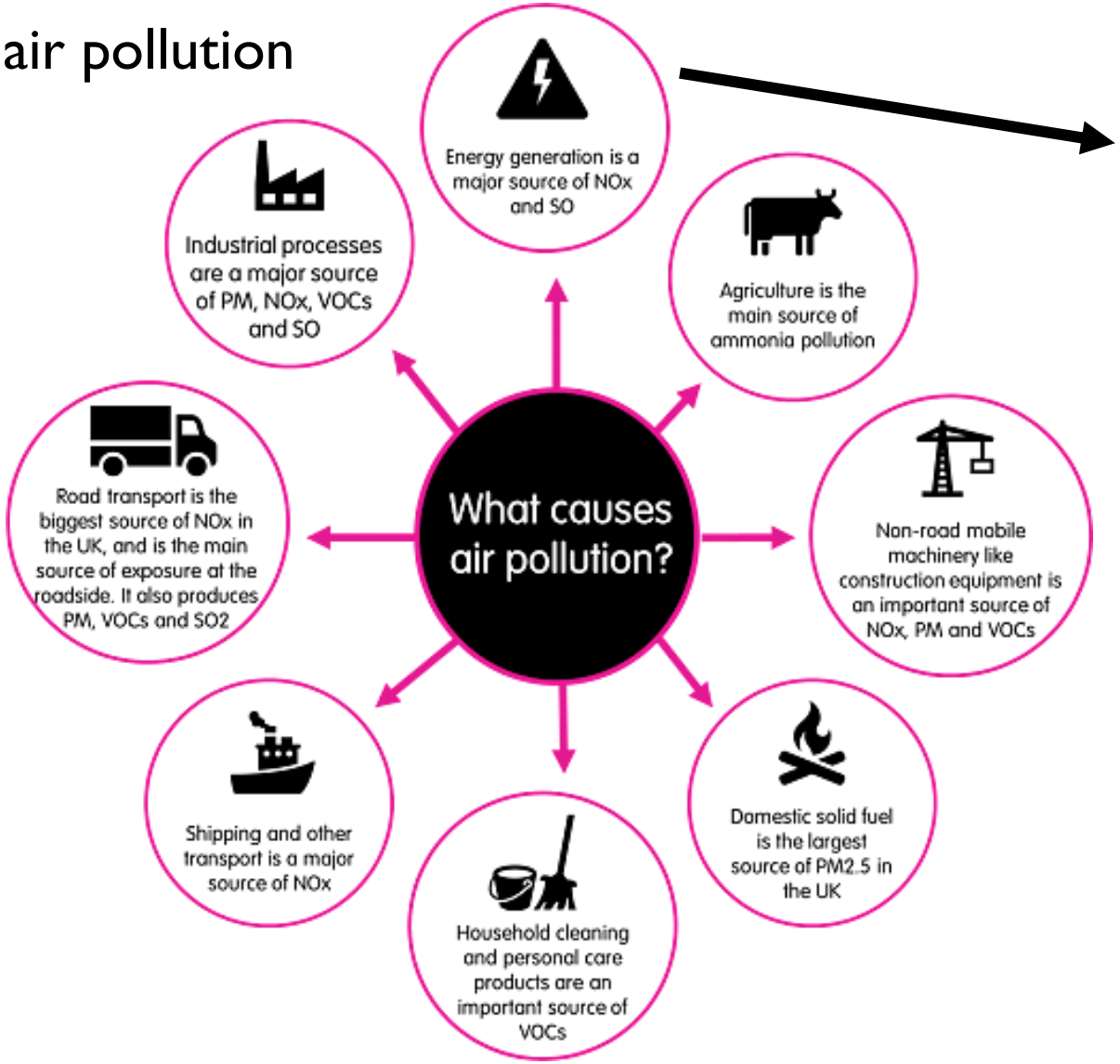
Air Pollution

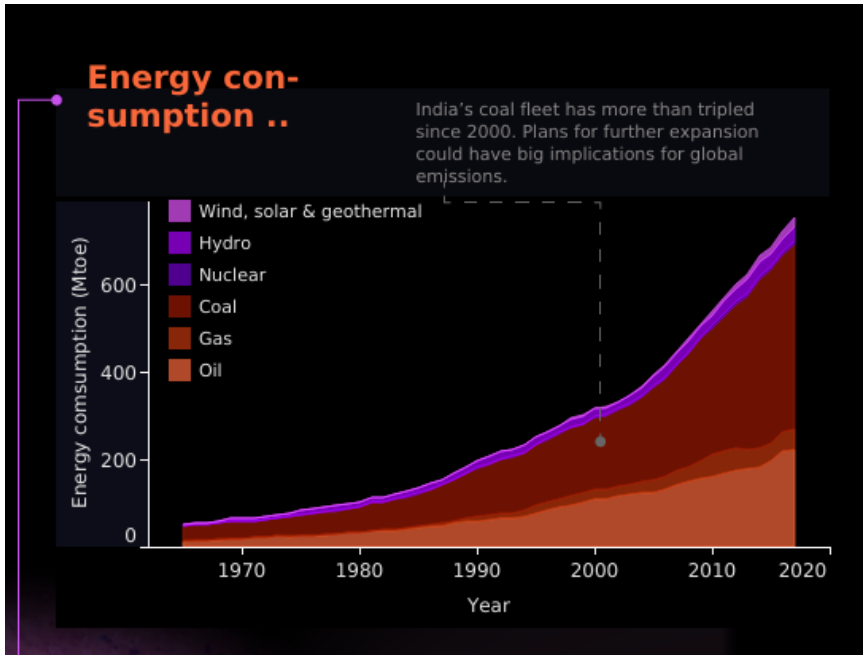
- Air pollution is defined as the release of pollutants into the atmosphere that are harmful to human health and the environment as a whole.
- According to the World Health Organization (WHO), air pollution kills approximately seven million people per year around the world.
- Nine out of ten people now breathe air that exceeds the WHO's pollution guideline levels.

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Bing Search, CCBY

Air Pollution

- Causes of air pollution





Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Image: https://www.carbonbrief.org/the-carbon-brief-profile-india

Classification of Pollutants: Origin

- Air Pollutants can be classified as
- ***(a) Primary Pollutants***
- ***(b) Secondary Pollutants***
- ***Primary Pollutants:-*** Primary pollutants are those which are directly emitted from identifiable source
- These pollutants are emitted directly to the atmosphere.
  - e.g.

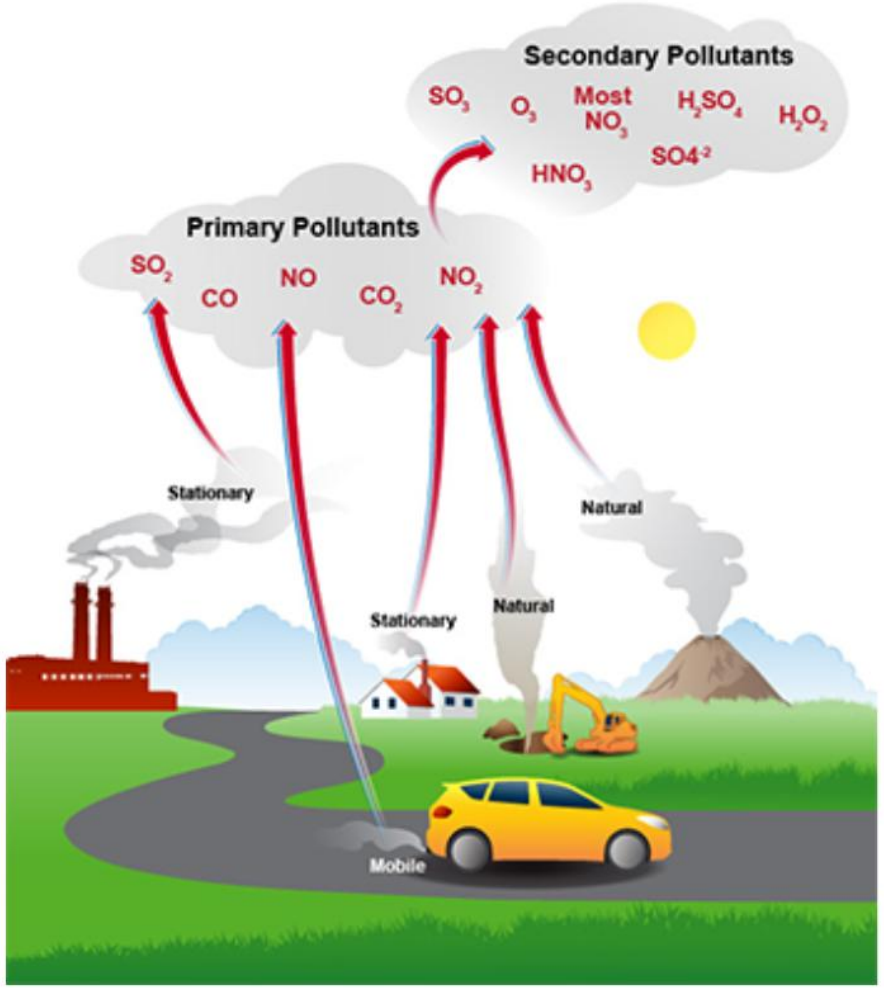
    (a) Particulate matter

    (b) Oxides of Sulphur

    (c) Oxides of Nitrogen

    (d) Carbon Monoxide

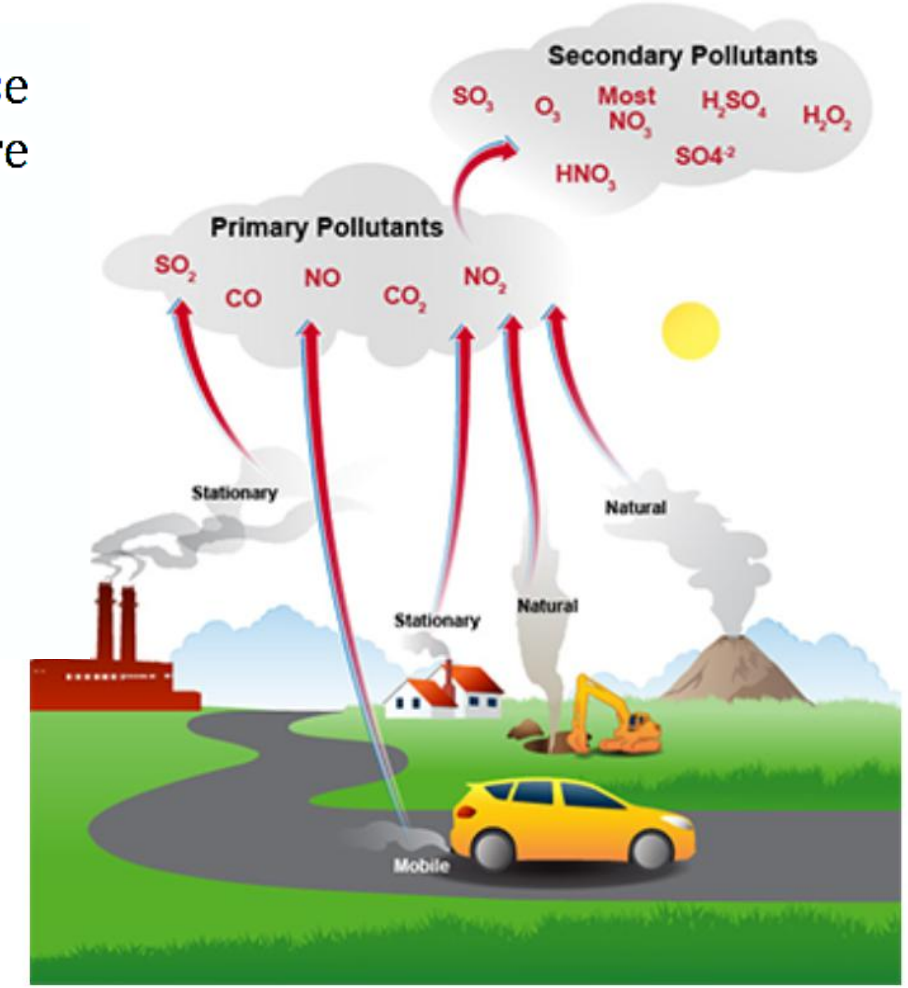
    (e) Radioactive Compounds



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Image: https://www.epa.gov/expobox/exposure-assessment-tools-media-air

Classification of Pollutants: Origin

- ***Secondary Air Pollutants:-*** Secondary air pollutants are those which are produced in the air by the reaction of two or more primary pollutants.
- e.g.
- Ozone
- Formaldehyde
- Photochemical smog
- Peroxy acetyl nitrate (PAN)



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Image: https://www.epa.gov/expobox/exposure-assessment-tools-media-air

Classification of Pollutants: Chemical Composition

(a) Organic Pollutants

(b) In Organic Pollutants

(a) Organic Pollutants:- Organic Pollutants contain carbon and hydrogen.

e.g.

- Hydro Carbon
- Aldehyde and Ketones
- Alcohols
- Organic Sulphur compounds

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

Classification of Pollutants: Chemical Composition

- ***Inorganic Pollutants:-*** Inorganic compounds mainly includes

(a) Carbon monoxide (CO)

(b) Carbon dioxide (CO_2)

(c)Carbonates

(d) Oxides of sulphur, nitrogen

(e) Ozone

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

Classification of Pollutants: State of Matter

- Air pollutants according to state of matter are classified as

(a) Natural Contaminants

(b) Particulate Matter

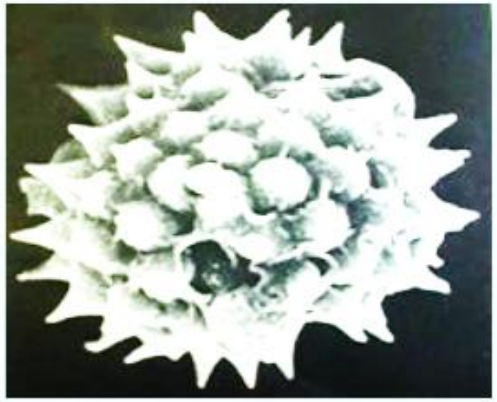
(c) Gases and Vapours

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

Classification of Pollutants: State of Matter

Natural Contaminants

- The air contaminants which are produced from natural sources are called natural contaminants
- Eg. Pollen grains are emitted from weeds, grasses and trees
- This pollen grains are irritating to some individuals
- Size of pollen grains = 10 to 50 (μ)



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

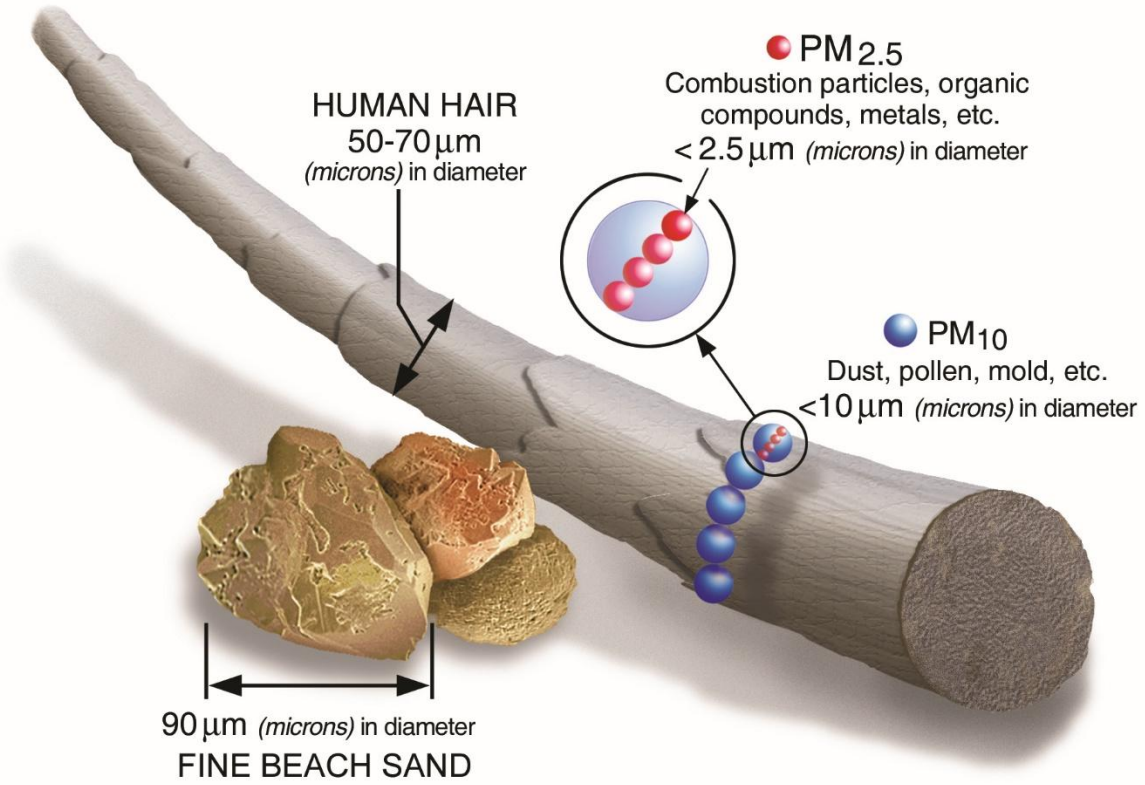
Classification of Pollutants: State of Matter

Particulate Matter

- These may be liquid or solid. The particulate matter are identified as aggregates which are larger than 0.002 μ but smaller than 500 μ

Dust (1 To 10,000 μ) : Small solid particles resulting from break up of large masses through processes such as crushing, grinding or blasting.

Smoke (0.5 To 1 μ) Consist of finely divided solid particles produced by incomplete combustion of organic particles such as coal, wood, or tobacco.



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

Classification of Pollutants: State of Matter

Particulate Matter

- ***Mist:-*** These are liquid particles formed by condensation of vapor
- ***Fog (1 to 40 μ):-*** High concentration of mist is called as fog. It is dispersion of liquid particles in air.
- ***Fumes (0.03 to 0.3 μ):-*** Fine solid particles formed by the condensation of vapors of solid material. The fumes are generally emit from melting operations.
- ***Ash (1 to 1000 μ):-*** fine, noncombustible particles are known as fly ash.
- ***Spray (10 to 1000 μ):-*** Liquid particles formed by automation

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

Classification of Pollutants: State of Matter

Gases and Vapors

- ***Carbon Monoxide:- (CO)*** This is an odorless, tasteless and colorless gas formed due to incomplete combustion of carbonaceous matter. The sources of carbon monoxide are motor vehicles, burning of fossil fuel and industrial processes.
- ***Oxides of Sulphur:-*** They Include 4 different gaseous compounds such as Sulfur monoxide (SO), Sulfur dioxide (SO_2), Sulphur trioxide (SO_3) and sulphur tetra oxide (SO_4).
- Sulphur dioxide is a colorless, non flammable gas with suffocating odor.

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

Classification of Pollutants: State of Matter

Gases and Vapors

- ***Oxides of Nitrogen Include 4 gaseous compound:-*** Nitric oxide (NO), Nitrogen dioxide (NO_2), Nitrous Oxide (N) Nitrogen penta oxide (N_2O_5)
- In air pollution, Nitric oxide (NO) and Nitrogen dioxide are of prime concern. The sources of this pollutants are Industrial Processes, automobile exhausts.

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

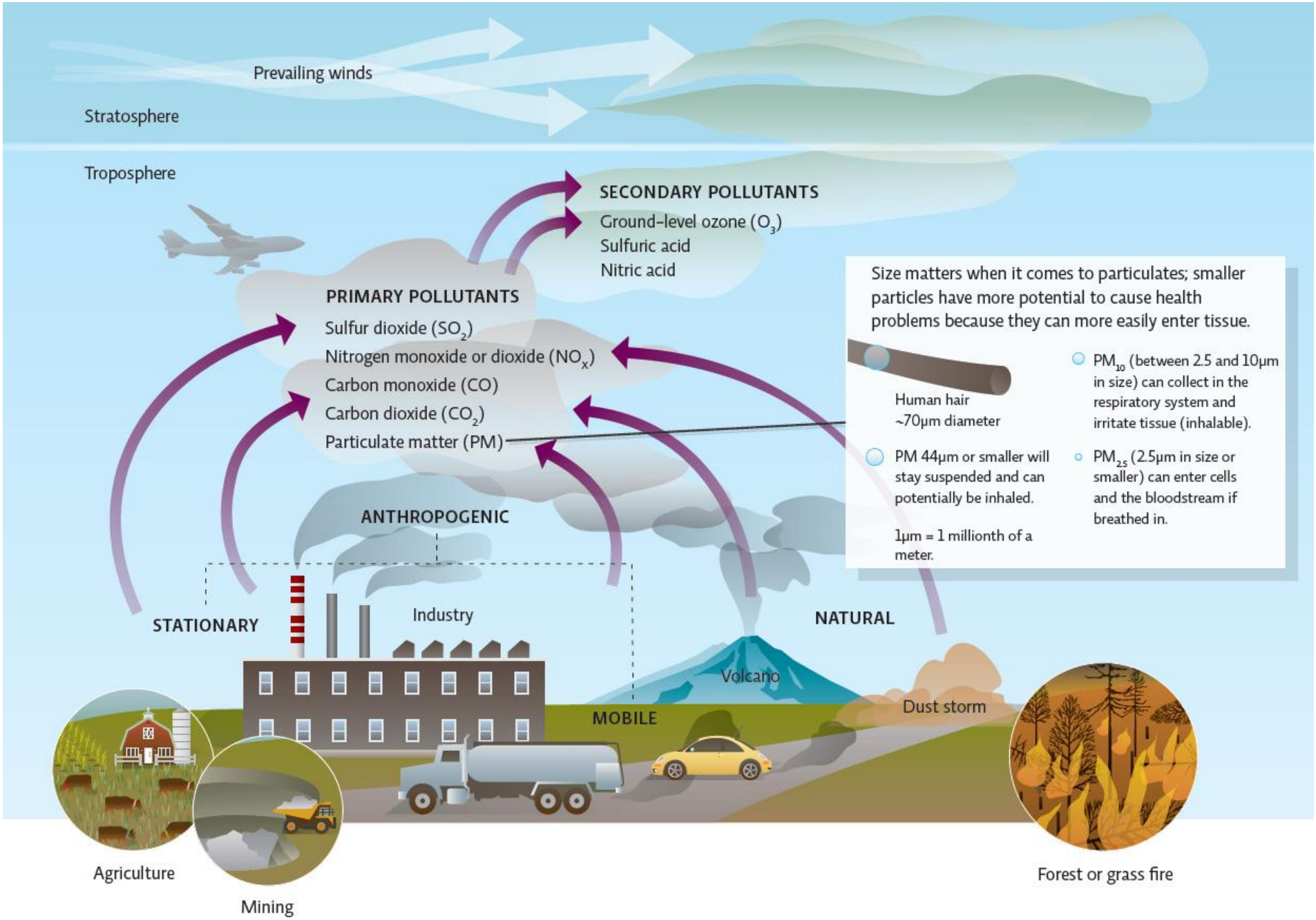
Classification of Pollutants: State of Matter

Gases and Vapors

- Organic compounds containing only carbon and hydrogen are classified as Hydrocarbons.
- They are of 2 types.
- ***(a) Aliphatic hydrocarbon***
- ***(b) Aromatic hydrocarbon***
- Hydrocarbon present in the atmosphere in the atmosphere from both natural and man made sources.
- The major sources of hydrocarbon is burning of gasoline and industrial processes.

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

Classification of Pollutants: Origin



Classification of Pollutants: State of Matter

Gases and Vapors

Major classes	Subclasses	Typical members of subclasses
Particulates	Solid	Dust, smoke, fumes, fly ash
	Liquid	Mist, spray
Gases		
Organic	Hydrocarbons	Hexane, benzene, ethylene, methane, butane, butadiene
	Aldehydes and ketones	Formaldehyde, acetone
	Other organics	Chlorinated hydrocarbons, alcohols
Inorganic	Oxides of carbon	Carbon monoxide, carbon dioxide
	Oxides of sulfur	Sulfur dioxide, sulfur trioxide
	Oxides of nitrogen	Nitrogen dioxide, nitric oxide
	Other inorganics	Hydrogen sulfide, hydrogen fluoride, ammonia

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985)

Thank you

For any questions, please feel free to reach out to me.